The Unit Circle and Exact Values

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Exact values

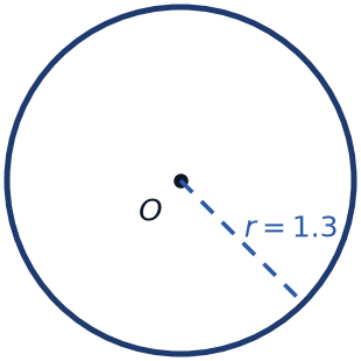
Write the exact value of each expression:

a $\sin\frac{\pi}{6}$ **b** $\cos\frac{3\pi}{4}$ **c** $\tan\frac{5\pi}{6}$ **d** $\sin\frac{3\pi}{2}$

2 [Foundational] • Target: Points on the unit circle

The terminal side of angle θ in standard position passes through the point $P\left(-\frac{3}{5}, \frac{4}{5}\right)$ on the unit circle.

Find:



a $\sin\theta$ **b** $\cos\theta$

c $\tan\theta$ **d** the quadrant containing θ

3 [Foundational] • Target: Points on the unit circle

State the reference angle for each angle:

a $\frac{5\pi}{6}$ **b** $\frac{4\pi}{3}$ **c** $-\frac{\pi}{4}$ **d** $\frac{7\pi}{2}$

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4 [Foundational] • Target: Points on the unit circle

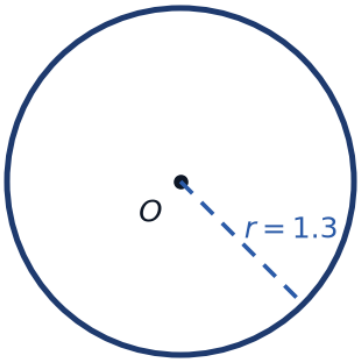
Identify the quadrant(s) in which each pair of conditions holds:

a $\sin\theta < 0$ and $\cos\theta > 0$

b $\tan\theta > 0$ and $\sin\theta < 0$

5 [Foundational] • Target: Points on the unit circle

Given that $\cos\theta = -\frac{5}{13}$ and θ is in Quadrant III, find $\sin\theta$ and $\tan\theta$ exactly.



6 [Foundational] • Target: Even/odd properties and periodicity

Use the fact that cosine is even and sine is odd to simplify each expression:

a $\sin(-\theta) + \sin\theta$

b $\cos(-\theta) - \cos\theta$

c $\frac{\sin(-\theta)}{\cos(-\theta)}$

7 [Foundational] • Target: Even/odd properties and periodicity

Use periodicity to find the exact value of each expression:

a $\sin\frac{13\pi}{6}$ **b** $\cos\frac{17\pi}{4}$ **c** $\tan\frac{9\pi}{4}$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Solving basic trigonometric equations on the unit circle

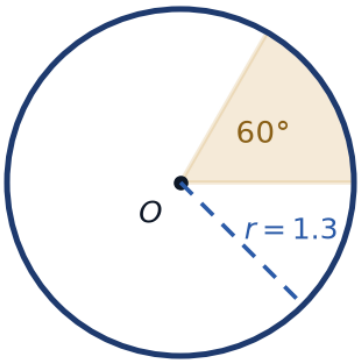
Find all solutions to $\sin\theta = -\frac{1}{2}$ in the interval $[0, 2\pi)$.

9 [Intermediate] • Target: Solving basic trigonometric equations on the unit circle

Find all solutions to $2\cos\theta + \sqrt{3} = 0$ in the interval $[0, 2\pi)$.

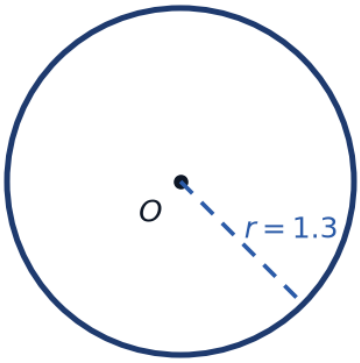
10 [Intermediate] • Target: Building the unit circle from special triangles

Using the 30-60-90 triangle, derive the exact coordinates of the point on the unit circle at $\theta = \frac{\pi}{3}$.



11 [Intermediate] • Target: Building the unit circle from special triangles

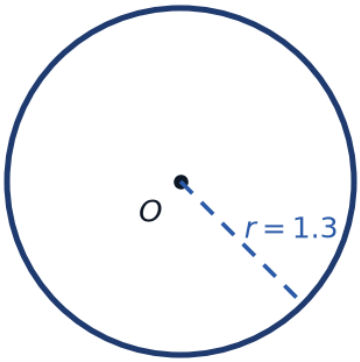
State the coordinates of all four points on the unit circle where the terminal side lies along an axis ($\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$).



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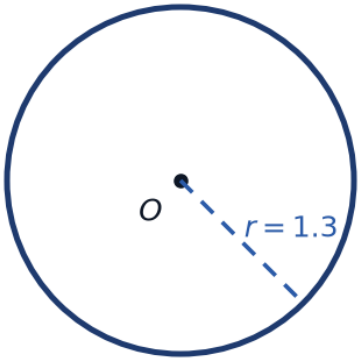
12 **[Intermediate]** • **Target: Using the Pythagorean identity with unit circle points**

A point on the unit circle has x-coordinate $\frac{7}{25}$ and lies in Quadrant IV. Find its y-coordinate using $x^2 + y^2 = 1$.



13 **[Intermediate]** • **Target: Using the Pythagorean identity with unit circle points**

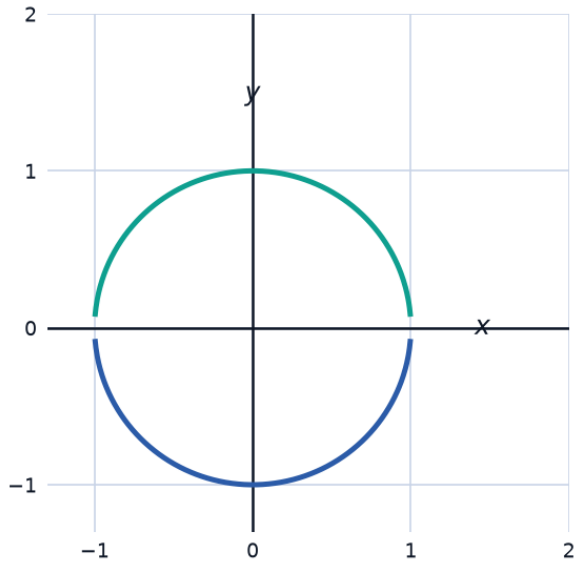
Given $\sin\theta = 0.6$ and θ in Quadrant II, find $\cos\theta$ and $\tan\theta$ exactly (using the fact $0.6 = \frac{3}{5}$).



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14 **[Intermediate]** • **Target: The unit circle picture**

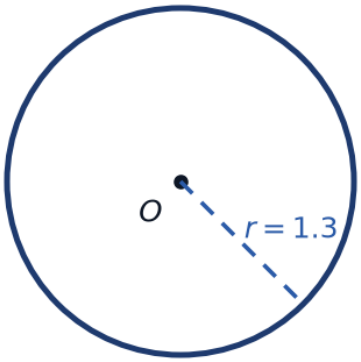
The unit circle $x^2 + y^2 = 1$ is shown. Use it to estimate $\cos\left(\frac{2\pi}{3}\right)$ and $\sin\left(\frac{2\pi}{3}\right)$ visually, then state the exact values.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: All six trig functions from a unit circle point** • **Cross-Lesson: 7.04 Applications: Growth and Decay**

A point on the unit circle has coordinates $\left(-\frac{5}{13}, \frac{12}{13}\right)$. Find all six trigonometric function values for the corresponding angle θ.



16 **[Challenging]** • **Target: Solving for the angle given exact coordinates** • **Cross-Lesson: 8.01 Rational Functions and Asymptotes**

Find all values of θ in $[0, 2\pi)$ for which the unit circle point is exactly $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$.

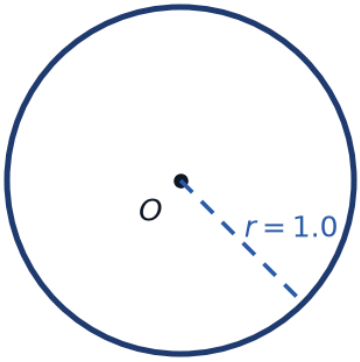
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17 [Challenging] • Target: Solving basic trigonometric equations on the unit circle • Cross-Lesson: 8.02 Graphing Rational Functions

Consider a point P on the unit circle such that its coordinates are $(\cos\theta, \sin\theta)$. Let α be an angle in the first quadrant and β be an angle in the third quadrant, with $\cos\alpha = \frac{3}{5}$ and $\sin\beta = -\frac{4}{5}$. If the angle θ is such that $\theta = \alpha + \beta$, find the value of $\cos\theta$.

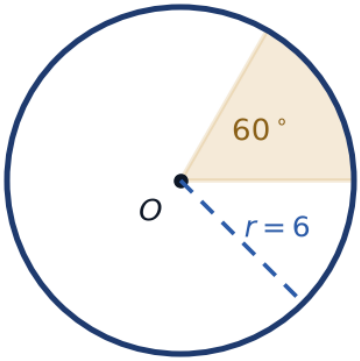
18 [Challenging] • Target: Building the unit circle from special triangles • Cross-Lesson: 8.03 Introduction to Limits

On the unit circle, a point P is located at coordinates (x, y). If the angle formed by the positive x-axis and the line segment from the origin to P is θ, what is the value of x in terms of θ?



19 [Challenging] • Target: Using the Pythagorean identity with unit circle points • Cross-Lesson: 9.01 Arithmetic Sequences and Series

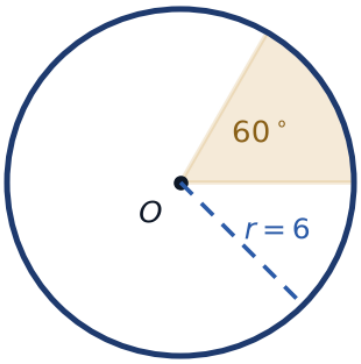
If $\sin\theta = \frac{3}{5}$ with θ terminating in Quadrant I, find the exact values of $\cos\theta$, $\tan\theta$, and $\sin(2\theta)$.



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20 [Challenging] • Target: The unit circle picture • Cross-Lesson: 9.02 Geometric Sequences and Series

Convert the polar coordinate point $P\left(6, \frac{4\pi}{3}\right)$ to exact Cartesian coordinates (x, y).



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Coordinates on the Unit Circle

On the unit circle, what do the coordinates (x, y) of a point at angle θ represent?

a. $x = \cos\theta$ and $y = \sin\theta$

b. $x = \sin\theta$ and $y = \cos\theta$

c. $x = \tan\theta$ and $y = \cot\theta$

d. $x = r\cos\theta$ where $r > 1$

22 [Foundational] • Target: Coordinates on the Unit Circle

Find the exact coordinates on the unit circle corresponding to $\theta = \frac{5\pi}{6}$.

a. $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

b. $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

c. $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

d. $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

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23 [Foundational] • Target: Coordinates on the Unit Circle

Find the exact coordinates on the unit circle for $\theta = \frac{4\pi}{3}$.

a. $\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$

b. $\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

c. $\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$

d. $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$

24 [Foundational] • Target: Reference Angles and Signs by Quadrant

Find the reference angle θ' for $\theta = 225^\circ$.

a. 45°

b. 35°

c. 135°

d. 55°

25 [Foundational] • Target: Reference Angles and Signs by Quadrant

In which quadrants is $\tan\theta$ positive?

a. Quadrants I and III

b. Quadrants I and II

c. Quadrants I and IV

d. Quadrants II and IV

26 [Foundational] • Target: Reference Angles and Signs by Quadrant

If $\cos\theta < 0$ and $\sin\theta > 0$, in which quadrant does the terminal ray of θ lie?

a. Quadrant II

b. Quadrant I

c. Quadrant III

d. Quadrant IV

27 [Foundational] • Target: Exact Values of the Six Trig Functions

Find the exact value of $\tan\left(\frac{2\pi}{3}\right)$.

a. $-\sqrt{3}$

b. $\sqrt{3}$

c. $-\frac{\sqrt{3}}{3}$

d. -1

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28 [Foundational] • Target: Exact Values of the Six Trig Functions

Find the exact value of $\csc\left(\frac{7\pi}{4}\right)$.

a. $-\sqrt{2}$

b. $\sqrt{2}$

c. $-\frac{\sqrt{2}}{2}$

d. -2

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Exact Values of the Six Trig Functions

Evaluate $\cos(3\pi) - \sin\left(\frac{3\pi}{2}\right)$.

a. 0

b. -2

c. 2

d. 1

30 [Intermediate] • Target: Quadrantal Angles and Negative Angles

Which of the following values is undefined?

a. $\tan(90^\circ)$

b. $\sin(90^\circ)$

c. $\cos(90^\circ)$

d. $\cot(45^\circ)$

31 [Intermediate] • Target: Quadrantal Angles and Negative Angles

Evaluate $\cos(-\pi/3)$.

a. $1/2$

b. $-1/2$

c. $\sqrt{3}/2$

d. $-\sqrt{3}/2$

32 [Intermediate] • Target: Quadrantal Angles and Negative Angles

Evaluate $\sin(-\pi/4)$.

a. $-\frac{\sqrt{2}}{2}$

b. $\frac{\sqrt{2}}{2}$

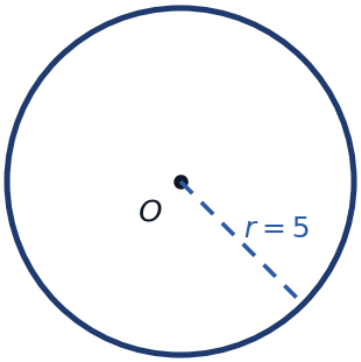
c. -1

d. 0

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33 [Intermediate] • Target: Quadrantal Angles and Negative Angles

Consider the unit circle shown, where a point $P(x, y)$ lies on the circle in the first quadrant. A ray from the origin through P forms an angle θ with the positive x-axis. If the y-coordinate of P is $\frac{\sqrt{3}}{2}$ and $\cos\theta = \frac{1}{2}$, find the value of x.



a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$

c. 1

d. $\frac{\sqrt{2}}{2}$

34 [Intermediate] • Target: Exact values

Consider a unit circle centered at the origin. A point $P(x, y)$ lies on the unit circle in the second quadrant such that the angle formed by the positive x-axis and the line segment OP is θ. If $\sin(\theta) = \frac{3}{5}$ and the y-coordinate of P is $y = \frac{3}{5}$, determine the x-coordinate of P.

a. $-\frac{4}{5}$

b. $\frac{4}{5}$

c. $-\frac{3}{5}$

d. $-\frac{25}{16}$

35 [Intermediate] • Target: Points on the unit circle

Consider a point P on the unit circle in the second quadrant such that the y-coordinate of P is $\frac{3}{5}$. Let θ be the angle in standard position whose terminal side passes through P. Find the value of $\cos\theta$.

a. $-\frac{3}{5}$

b. $\frac{3}{5}$

c. $-\frac{4}{5}$

d. $\frac{4}{5}$

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36 [Intermediate] • Target: Even/odd properties and periodicity

Consider a point $P(\cos\theta, \sin\theta)$ on the unit circle in the Cartesian plane. If the distance from the origin to the point $Q(x, y)$ is 5, and the angle between the positive x-axis and the line segment OQ is α, such that $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. If $x = \cos\alpha$ and $y = \sin\alpha$, find the maximum possible value of the product of the x-coordinates of P and Q, i.e., $\cos\theta \cdot \cos\alpha$.

a. 4/5

b. 3/5

c. 1

d. 5/3

37 [Intermediate] • Target: Solving basic trigonometric equations on the unit circle

Consider a point P on the unit circle in standard position. If the angle θ in standard position has a terminal side passing through P such that $\cos(\theta) = \frac{1}{2}$ and $\sin(\theta) > 0$, find the value of $\sin(\theta)$.

a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$

c. $-\frac{\sqrt{3}}{2}$

d. -1

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Building the unit circle from special triangles • Cross-Lesson: 8.03 Introduction to Limits

On the unit circle, a point $P(x, y)$ corresponds to an angle θ in standard position. If the angle $\theta = \frac{\pi}{3}$, what is the value of the y-coordinate of point P?

a. $\frac{1}{2}$

b. $\frac{\sqrt{2}}{2}$

c. $\frac{\sqrt{3}}{2}$

d. 1

39 [Challenging] • Target: Using the Pythagorean identity with unit circle points • Cross-Lesson: 9.01 Arithmetic Sequences and Series

If $\cos\theta = \frac{60}{61}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{60}{61}$

b. $\frac{11}{61}$

c. $\frac{171}{610}$

d. $-\frac{11}{61}$

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40 [Challenging] • Target: The unit circle picture • Cross-Lesson: 9.02 Geometric Sequences and Series

If $\cos\theta = \frac{35}{37}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{157}{370}$

b. $-\frac{12}{37}$

c. $\frac{35}{37}$

d. $\frac{12}{37}$

41 [Challenging] • Target: All six trig functions from a unit circle point • Cross-Lesson: 9.03 The Binomial Theorem

If $\cos\theta = \frac{45}{53}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{28}{53}$

b. $-\frac{28}{53}$

c. $\frac{333}{530}$

d. $\frac{45}{53}$

42 [Challenging] • Target: Solving for the angle given exact coordinates • Cross-Lesson: 1.01 Functions and Function Notation

If $\cos\theta = \frac{56}{65}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{79}{130}$

b. $\frac{33}{65}$

c. $-\frac{33}{65}$

d. $\frac{56}{65}$

43 [Challenging] • Target: Coordinates on the Unit Circle • Cross-Lesson: 1.02 Domain and Range

If $\cos\theta = \frac{63}{65}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $-\frac{16}{65}$

b. $\frac{16}{65}$

c. $\frac{9}{26}$

d. $\frac{63}{65}$

44 [Challenging] • Target: Reference Angles and Signs by Quadrant • Cross-Lesson: 1.03 Average Rate of Change

If $\cos\theta = \frac{55}{73}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{553}{730}$

b. $\frac{55}{73}$

c. $-\frac{48}{73}$

d. $\frac{48}{73}$

45 [Challenging] • Target: Exact Values of the Six Trig Functions • Cross-Lesson: 2.01 Right-Triangle Trigonometry

If $\cos\theta = \frac{84}{85}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{13}{85}$

b. $\frac{84}{85}$

c. $-\frac{13}{85}$

d. $\frac{43}{170}$

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